National University of Singapore

Institute of Systems Science
Intelligent Reasoning Systems
Project report

GROUP PROJECT REPORT · SHORT VERSION

AnchorAI

Explainable, budget-aware retention planning for gyms

Churn prediction · uplift estimation · rule-based reasoning · case-based reasoning
genetic algorithm with an exact CP-SAT benchmark · knowledge graph · grounded staff assistant

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Abstract

About one gym member in three leaves each year, and most stop attending weeks before they cancel. Retention tools flag members at risk but leave the decision that matters (whom to contact this week, with which offer, within a fixed budget and limited staff hours) to guesswork. AnchorAI makes that decision and explains it. A calibrated ensemble predicts churn on real members; a forward-chaining rule base encodes the gym's playbooks and guardrails; case-based reasoning estimates what each offer would achieve for each member; a genetic algorithm, benchmarked against an exact CP-SAT solver, builds the weekly plan; a knowledge graph adds signals the churn model cannot see; and a grounded assistant answers staff questions from the system's own evidence. Segmentation and churn are evaluated on 4,000 real members of a public dataset; decisions on a documented synthetic scenario layer whose true offer effects are known, which shows that the method works when effects are of this kind, not how large they are in a real gym.

0.908
churn F1 on 800 unseen real members (PR-AUC 0.973)

+22–138%
more members kept than contacting by risk or by schedule, at equal budget

0.24%
largest gap of the genetic algorithm to the proven optimum

96.2%
of the assistant's judged sentences supported by its evidence

Contents

1 Introduction 1
2 System design 2
3 Data and the scenario layer 3
4 Methods 4
5 Implementation 7
6 Experiments and findings 7
7 Conclusions 12

References **13**

1 Introduction

Gym members rarely cancel on impulse: they stop coming first and cancel weeks later, and that lag is the window in which a well-chosen contact can still keep them. The median member retention rate was 66.4% in 2024 across 175 operators [1], and new members are the most fragile: in one fitness centre 63% stopped before their third month [2]. Current practice produces a list or a score but not a decision. Calendar campaigns spend on members who would have stayed, threshold rules ignore trends, and vendor risk scores give staff a number they cannot interrogate. The proposal's market research found that no vendor publicly exposes constrained allocation, that vendors target risk although targeting by expected response reduces churn more [3], and that the intelligence layer is priced for chains of five or more clubs. AnchorAI is therefore positioned as an *explainable, budget-aware retention planner for operators with one to five clubs.* No real operator was involved: the interviews that shaped the rules were simulated, and every comparison is against baseline policies in our own experiments.

Table 1 Objectives, how each is tested, and the verdicts argued in Section 6.

	Objective	Test and baseline	Verdict
O1	know who is at risk, and why	E1 segments (every k); E2 churn (visit rule, logistic regression); E7 graph signals	● Met
O2	know whom a contact would keep	E3 uplift (T-learner, risk, random)	● Partly met
O3	plan the week within budget, staff and stock	E4 plan value (contact-all, risk-top-k); E5 optimality and speed (CP-SAT)	● Met
O4	keep people in control	E6 rules and guardrails; tests of explanation and audit coverage	● Met
O5	usable by staff in plain language	E8 assistant (keyword router)	● Partly met
O6	reproducible and dependable	fresh rebuilds; tests with services down	● Met

2 System design

One web application serves administrators, managers and staff; one FastAPI service checks every request's role, audits every change and runs long jobs in the background; and three reasoning components do the work (Figure 1). They meet at three contracts: *data tables* written once by the data build and read by everything else; an *allowed-offer list* from the rule engine, so the optimiser never sees an offer the rules block; and an *explanation record* per plan item (drivers, rule trace, graph signals, ten similar past cases), which the plan's 'why' panel and the assistant both read, so they always give the same reason. Each week runs from scores to an approved, assigned call list, and logged outcomes feed the next week (Figure 2).

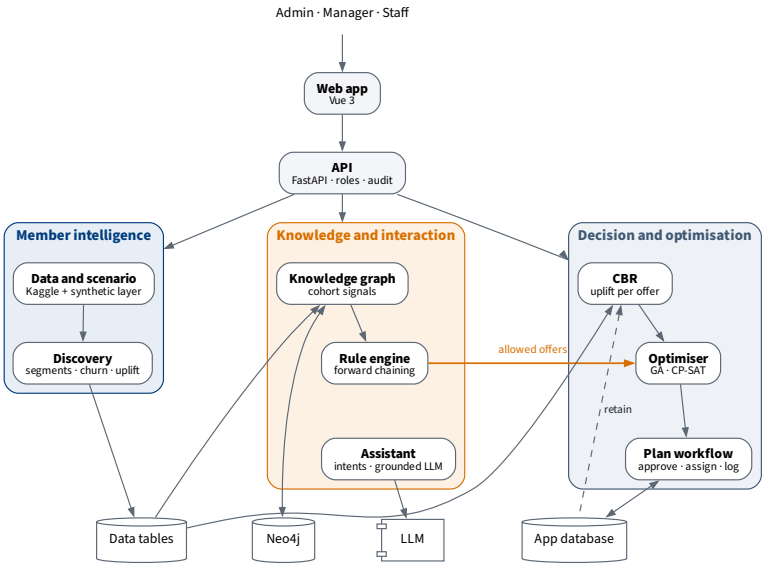


Figure 1 Architecture. Arrows show data flow; the dashed arrow is the learning loop through which logged outcomes return to the case base.

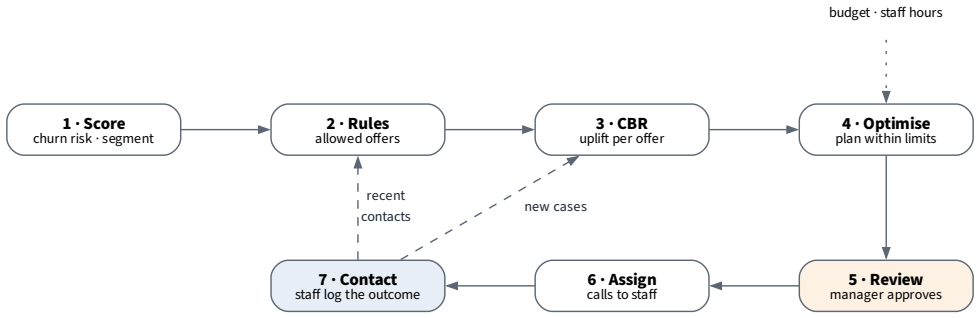


Figure 2 The weekly cycle: steps 1–4 run automatically within the week's budget and staff hours; the manager reviews and approves, and staff contact members and log outcomes.

3 Data and the scenario layer

Segmentation and churn prediction use only the 4,000 real members of the public *Gym Customers Features and Churn* dataset [4]: 14 columns, 26.5% churners, split 3,200/800 with stratification, and four derived features such as the visit trend (this month's visits over the lifetime rate), whose median is 0.69 for members who left and 1.00 for those who stayed (Figure 3). Two same-month columns may leak the label, so E2 reports the model with and without them. The snapshot has no clubs, classes, offers or contact history, so a documented *scenario layer* attaches a synthetic gym: three clubs, nine instructors and 18 classes consistent with each member's real visits; four offers with costs, staff minutes and stock; a heterogeneous ground-truth uplift for all 16,000 member–offer pairs (mean +5.8 points); two gym events (an instructor leaving and a class cancelled); and a randomised history of 2,400 past contacts (Figure 4). The ground truth is used only to grade, never by an estimator or planner. One command rebuilds every table, model and score in about 20 seconds from one seed.

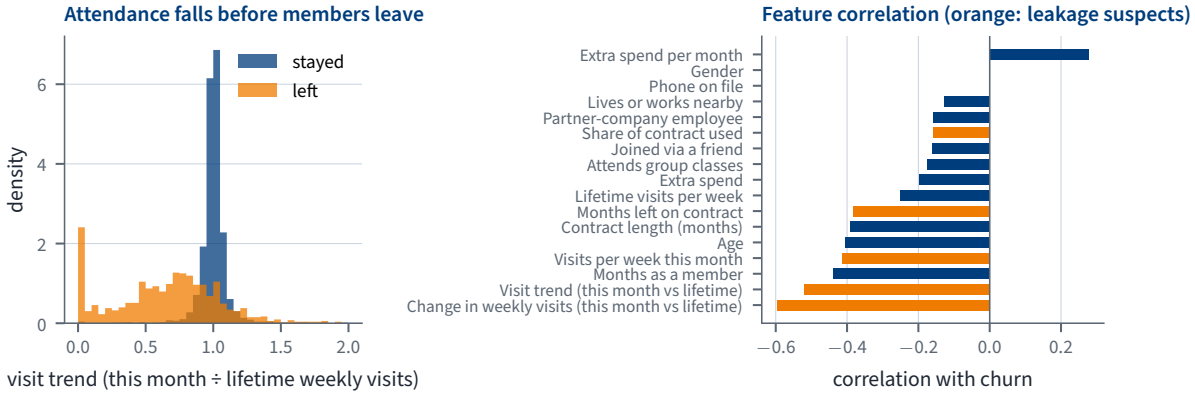


Figure 3 Left: the visit trend of members who stayed clusters at 1, while members who left spread below it, with a spike at 0 for those who did not come this month (medians 1.00 and 0.69). Right: correlation of each feature with churn; the two strongest are leakage suspects. Real Kaggle data.

Source: `scripts/figures.py` (`trend`), as in `a_features_leakage.ipynb`, cells 2–4.

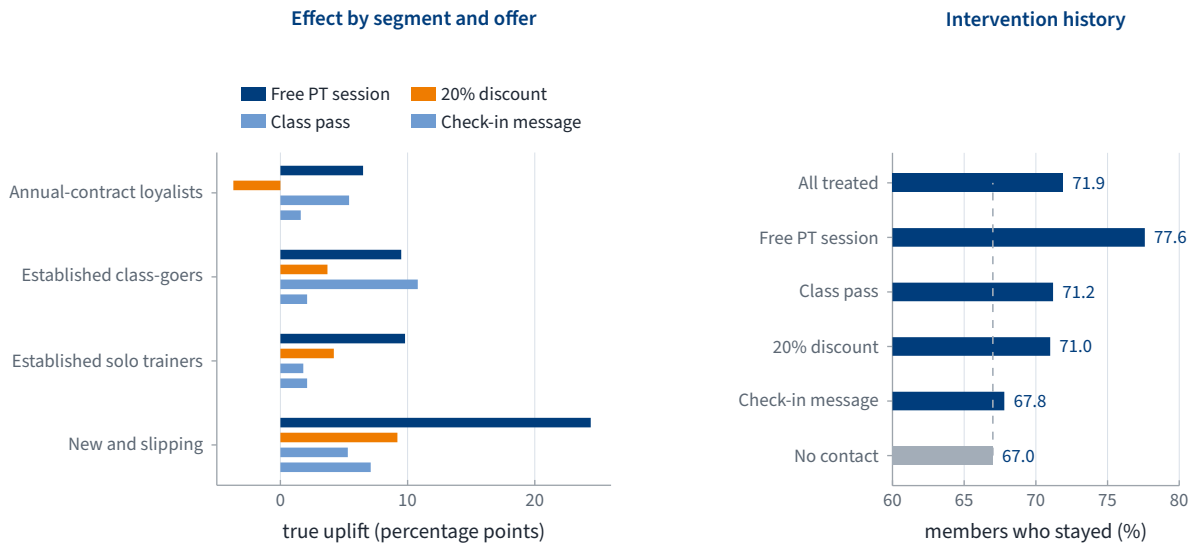


Figure 4 The scenario's ground truth. Left: mean true uplift by segment and offer; a discount backfires on annual members. Right: share who stayed in each arm of the randomised history against the 984 controls (dashed).

Source: `results/a5_true_uplift_by_segment.csv`; `a_scenario_uplift.ipynb`.

4 Methods

4.1 Member intelligence

K-means [5] on seven behavioural features gives segments that staff can name; k is chosen by a fixed rule on the silhouette [6] and names come from rules on the centres, so they survive a rebuild. Churn is predicted by the mean of a class-weighted logistic regression and LightGBM [7] over 17 features (Figure 5). Class weights matched SMOTE-NC [8] in cross-validation and were kept as simpler; a Platt calibrator [9] fitted on a held-out slice makes the probabilities honest, which matters because the planner multiplies them by uplift and value. Training members are scored out of fold, and each member's five drivers average the logistic-regression term with LightGBM's TreeSHAP value [10] (Figure 6). A T-learner [11], one outcome model per offer and one for controls, is the uplift benchmark.

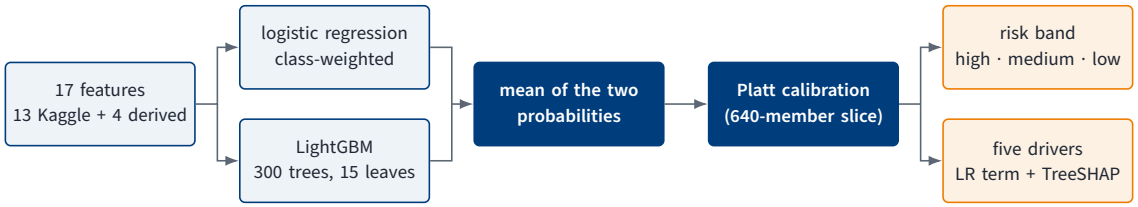


Figure 5 The churn pipeline, fitted on 2,560 training members and calibrated on the remaining 640.

4.2 Decision automation

A forward-chaining engine [12] runs 29 YAML rules: 8 *derive* rules name patterns (falling regular, new member, routine disrupted), 16 *playbooks* propose an offer on a channel, with twins for members without a phone, and 5 *guardrails* block offers (contact and offer caps, eligibility, no paid offers at low risk, no repeat training session). Conflicts resolve by salience, specificity and order (Figure 7); each rule fires once, and blocks are applied after the loop, so guardrails always win. Every rule cites its source, a validator rejects inconsistent rule bases, and administrators edit, test, version and roll back rules in the

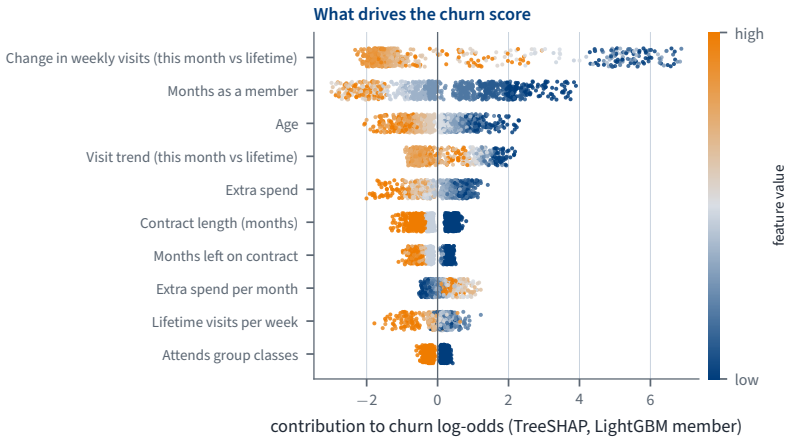


Figure 6 TreeSHAP values of the LightGBM member on the 800 test members, for the ten most influential features. Each dot is a member, coloured by its feature value; dots to the right push towards churn. Real Kaggle data.

Source: `scripts/figures.py` (churn), as in `a_churn.ipynb`, cell 5.

application. Thresholds were drafted on the training data and adjusted after three *simulated* manager interviews, which are disclosed as plausible opinion, not evidence.

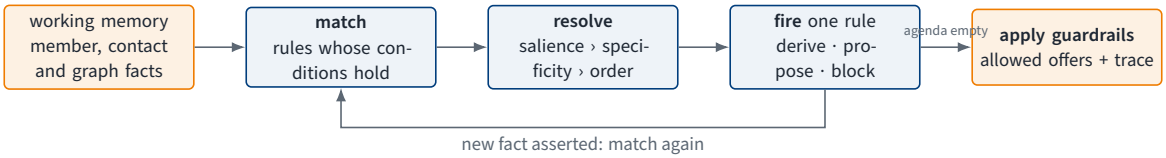


Figure 7 The recognise–act cycle. Each rule fires at most once (refraction); blocked offers are removed after the loop, so a guardrail always overrides a playbook.

4.3 Decision and optimisation

Each allowed member–offer–channel option is valued as $v_{mo} = \hat{u}_{mo} f_m H - c_o$, the estimated uplift [13] times the monthly fee over a cautious six-month horizon, minus the offer's cost. The uplift comes from case-based reasoning [14] (Figure 8): retrieve the 200 nearest past cases in one shared neighbourhood of treated and untreated members, reuse it as a local A/B test, revise by empirical-Bayes shrinkage towards the offer's average [15], and retain logged outcomes as evidence but never in the averages, because the planner's own choices would bias them. A null test with shuffled offers rejected an earlier design that measured risk instead of effect. The plan maximises total value within budget, staff minutes, trainer minutes and stock, plus two effects between members who train together: *spillover* and *diminishing returns* per instructor. These make the problem non-linear, so a genetic algorithm (DEAP [16]) searches it with a repair operator and a greedy seed, while CP-SAT [17] solves the linear case exactly and benchmarks the full one on small instances. A policy evaluator grades plans against the true effects and is never used to choose one.

4.4 Knowledge graph and staff assistant

A Neo4j knowledge graph [18] links members, classes, instructors, clubs, offers and events through read-only parameterised Cypher. It flags members whose anchor instructor left or whose class was cancelled, as rule facts cached so the rules work without Neo4j; it gives the optimiser its class cohorts; and it simulates an event's impact without changing data (Figure 10). The staff assistant routes a

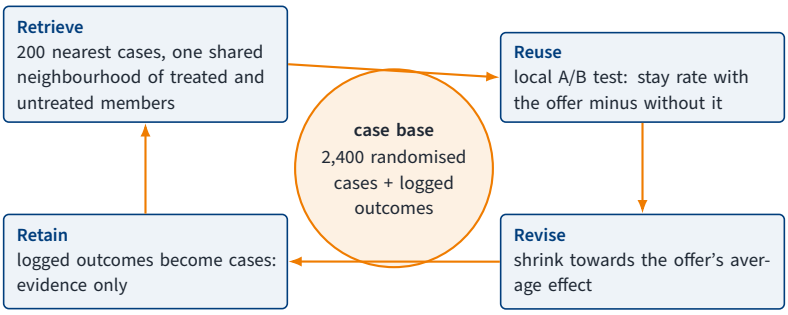


Figure 8 The case-based reasoner. The neighbours that produce an estimate are shown to staff as its precedents.

question to one of seven intents with sentence-embedding centroids [19] and an 'unknown' threshold, fills slots, runs one deterministic tool, and lets the language model only phrase the numbered evidence with citations, or answer that it does not know (Figures 9 and 11). Without the model, it returns the cited evidence itself.

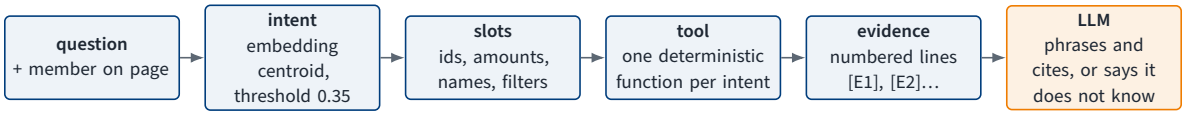


Figure 9 The assistant pipeline: the language model never chooses a tool and never sees data a tool did not return.

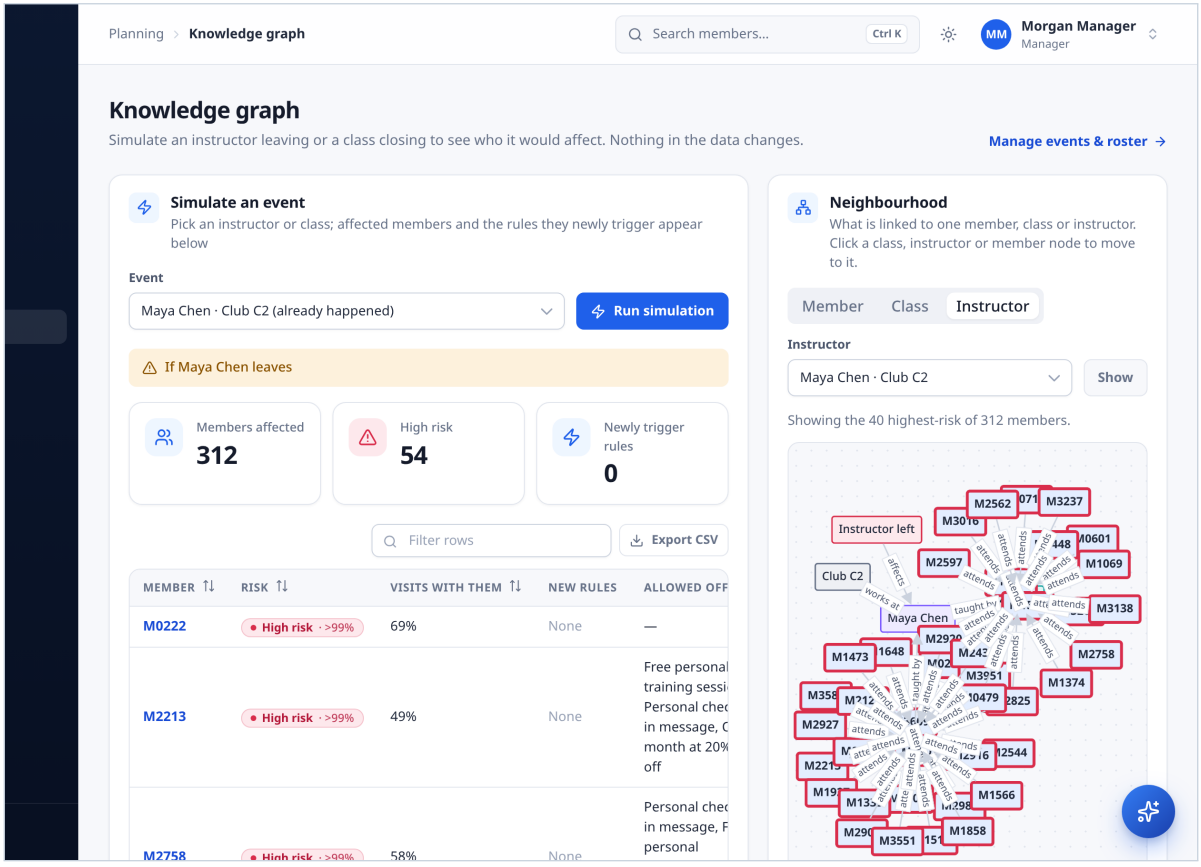


Figure 10 Simulating the departure of instructor Maya Chen in the graph explorer: 312 members are affected, 54 of them at high risk. Right: her neighbourhood, with the 40 highest-risk of her 312 members drawn.

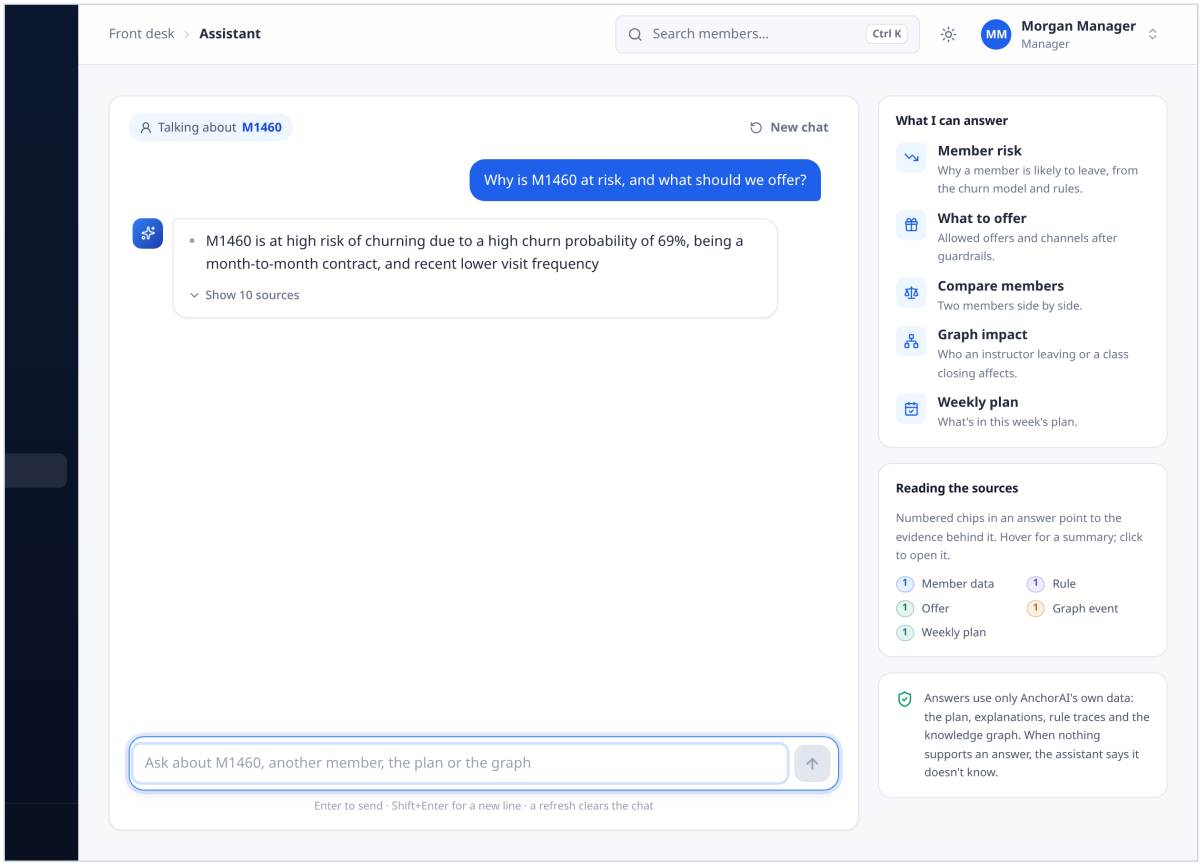


Figure 11 The assistant answering a manager about member M1460 with the local `qwen2.5:7b-instruct` model. The answer cites its sources, which open as evidence chips linking to the member, rule or case.

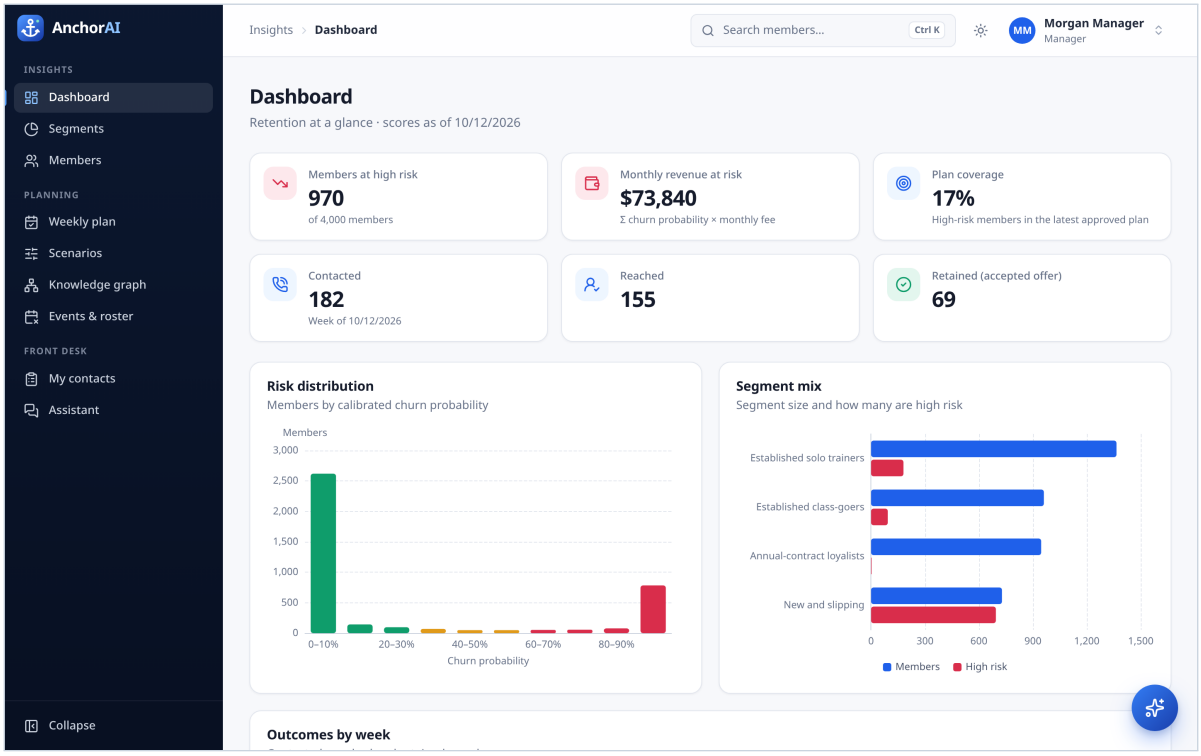
5 Implementation

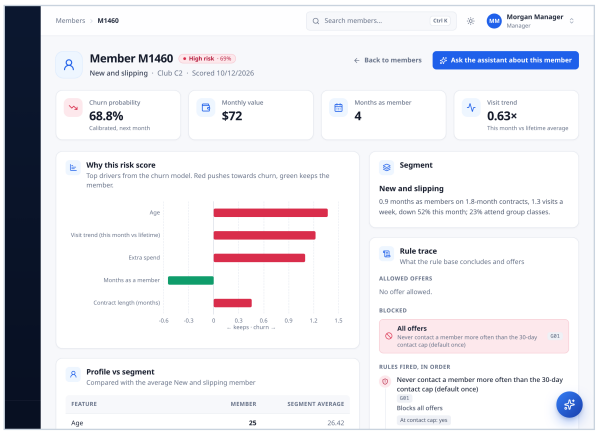
One command builds the data and models, migrates and seeds the database with a demo week, and loads the graph. Sixteen pages serve three roles (Figure 12): managers work from the dashboard, plan and scenarios; staff from a call sheet with talking points, message drafts and two-tap outcome logging; administrators from versioned rules, offers, data and models, users and an audit log of every write. Member 360 shows every component's reasoning about one member on one screen. Risk is always shown as text and colour; planning continues when Neo4j or the language model is down; and 195 offline tests guard against leakage, infeasible plans, rule order, unsafe queries and role or audit errors.

6 Experiments and findings

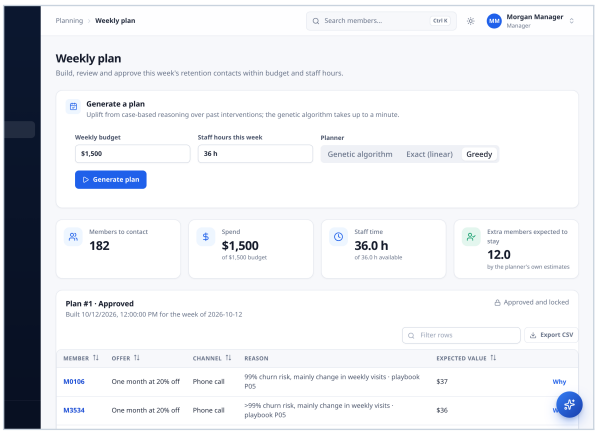
E1 and E2 use only real members; E3–E7 are graded by the scenario's ground truth, which no estimator or planner reads; E8 evaluates the assistant on the system's own evidence (Table 2).

Segments and churn (E1, E2). Silhouette is low and flat on behavioural data, but the four segments are stable across seeds and separate churn by a factor of almost 70 (Figure 13); the 728 'new and slipping' members hold about two-thirds of all churners. The calibrated ensemble beats both baselines on every metric (Figure 14); each risk band churns at about its predicted rate (high band 93.9% against 95.1% predicted), and without the two suspect same-month columns F1 falls to 0.817, the honest lower bound.





(a) Member 360: risk, drivers, profile and rule trace for one member.



(b) The weekly plan: 182 approved contacts, each with a reason.

Figure 12 The application on the demo data. Top: the manager's dashboard.

Recall for the rare churners aged 31 and over is 0.52, which we report with a mitigation rather than hide.

Uplift and plans (E3–E5). Case-based reasoning matches the T-learner overall and is better where effects are large (Figure 15), but cannot rank members for the small-effect check-in and class pass, which costs 3.7 members a week at the default 36 staff hours (21.0 against 24.6 extra members; level or ahead with more staff time). Planning by effect keeps 1.9 to 3.5 times as many members as contacting by schedule or risk with perfect knowledge (Figure 16) and 22–138% more on estimates. On noisy CBR estimates, however, the genetic algorithm keeps fewer members than greedy: the optimiser's curse [20], where stronger search selects estimates that happen to be too high. On its own objective the algorithm is near-optimal (0.8–2.3% from CP-SAT's optimum on the linear problem; within 0.24% on small full

Table 2 The eight experiments: real-data experiments use the 4,000 Kaggle members; scenario experiments are graded against the synthetic ground truth.

	Question	Key result	Verdict
E1	meaningful segments? REAL DATA	$k = 4$ by rule; stability ARI 1.00; churn from 94.6% to 1.4%	● Met
E2	churn model beats simple rules? REAL DATA	F1 0.908, PR-AUC 0.973, Brier 0.035 vs 0.862 / 0.953 / 0.056 (logistic) and 0.549 / 0.390 / 0.336 (visit rule)	● Met
E3	who is persuadable? SCENARIO LAYER	CBR ties the T-learner (Spearman 0.39 each), better for PT and discount	● Partly met
E4	more members per dollar? SCENARIO LAYER	22–138% more members kept than by risk or schedule; 0 of 198 plans infeasible	● Met
E5	GA good and fast enough? SCENARIO LAYER	within 0.24% of the optimum on small instances; 33 s for 2,000 members	● Partly met
E6	rules correct; guardrails matter? SCENARIO LAYER	0 validator issues; 29 of 29 cases; guardrails remove 31.3% of proposals	● Met
E7	graph signals add information? SCENARIO LAYER	no ranking gain; 243 extra low-risk members reached; +7 members at 120 h	● Partly met
E8	assistant accurate, grounded? SCENARIO LAYER	96.2% of sentences supported; held-out routing 50.0% vs 54.2% for keywords	● Partly met

Source: `SystemCode/notebooks/results/`; full method and caveats per experiment in the full report.

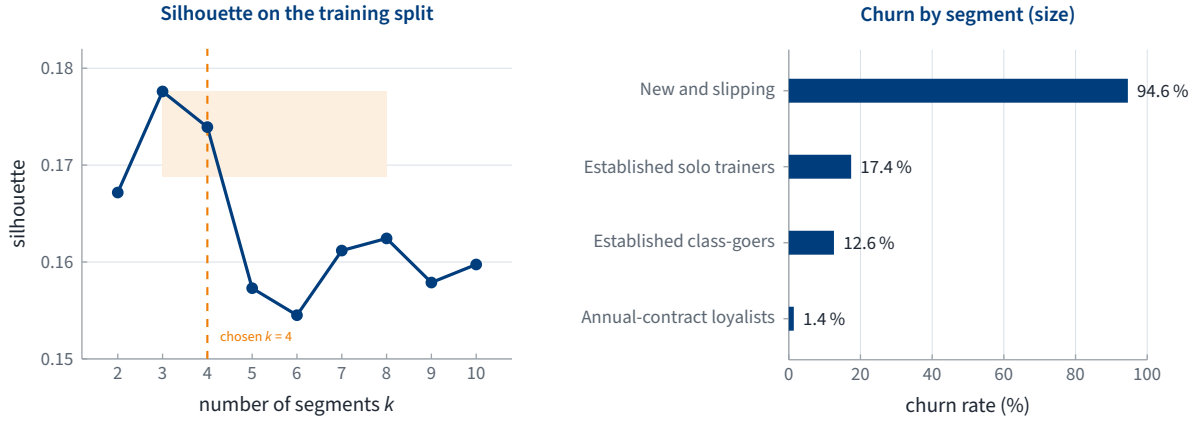


Figure 13 E1. Left: silhouette is low and nearly flat; the shaded band is within 5% of the best k between 3 and 8, and the rule picks the most detailed k inside it. Right: the four segments separate churn from 94.6% to 1.4% (sizes 728, 1,365, 961 and 946).
Source: `results/e1_silhouette_by_k.csv`, `results/e1_segment_profiles.csv`; `a_segmentation.ipynb`.

instances) and robust to the assumed spillover and diminishing-returns strengths (Figure 17), but CP-SAT is the better tool for the linear problem.

Rules, graph and assistant (E6–E8). The rule base is consistent, and its guardrails remove 31.3% of proposals, almost all repeat contacts within 30 days, at little cost where staff time binds (Figure 18); the scenario has no fatigue effect, so we claim safety, not retention. The graph's flags recover the designed event effects (hazard ×1.80 and ×1.32) but do not improve the risk ranking (Figure 19); they route 243 low-risk members with a disrupted routine to a free check-in, worth about 7 members when staff time allows. The assistant states only what its evidence supports (no invented number; 15 of 17 answers fully supported), but on 24 questions written by team members who never saw its training examples it

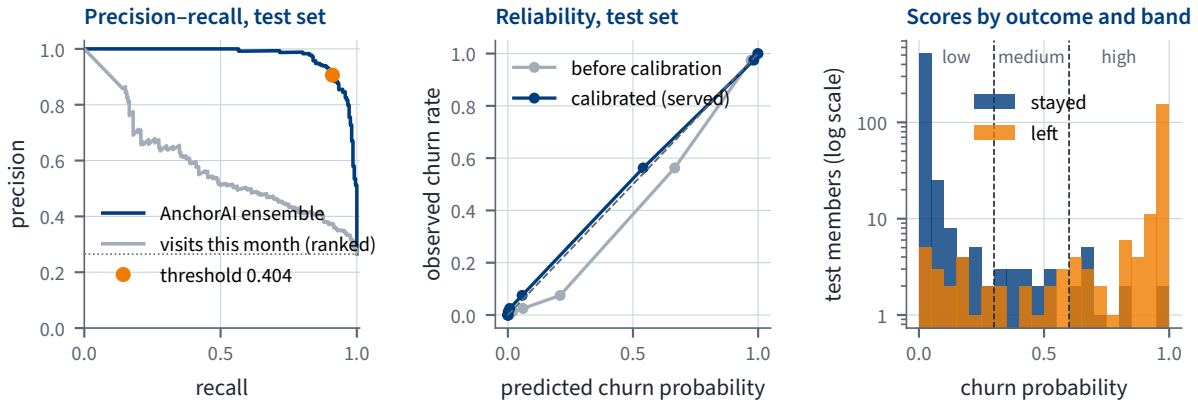


Figure 14 E2 on the 800 test members: precision–recall against ranking by this month's visits, reliability before and after calibration, and scores by outcome with the band cut-offs.

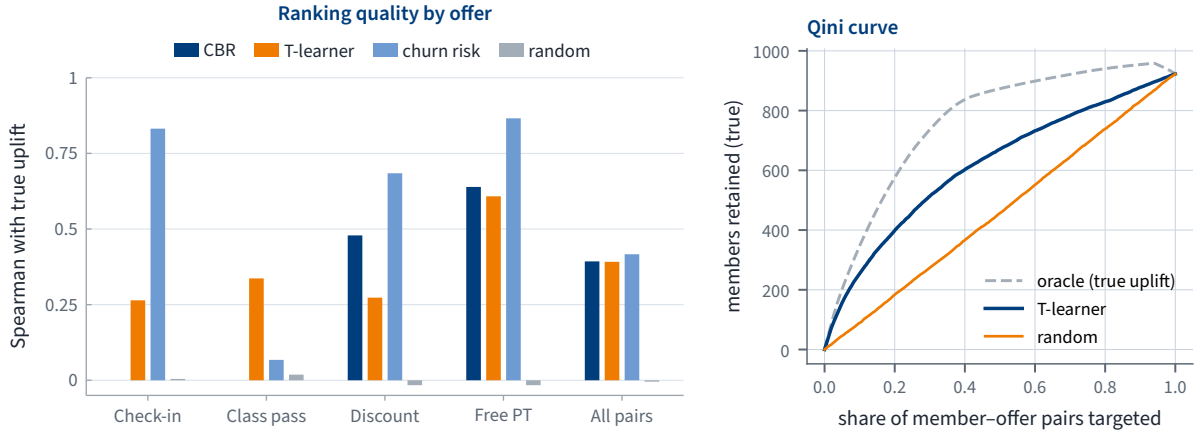


Figure 15 E3. Left: rank correlation between estimated and true uplift; CBR gives every member the offer's average for the check-in and the class pass, so it has no ranking there. Right: Qini curve of the T-learner against the oracle and random targeting.

Source: `results/e3_uplift_vs_truth.csv`, `results/a6_tlearner_recovery.csv`; `b_cbr.ipynb`, `a_scenario_uplift.ipynb`.

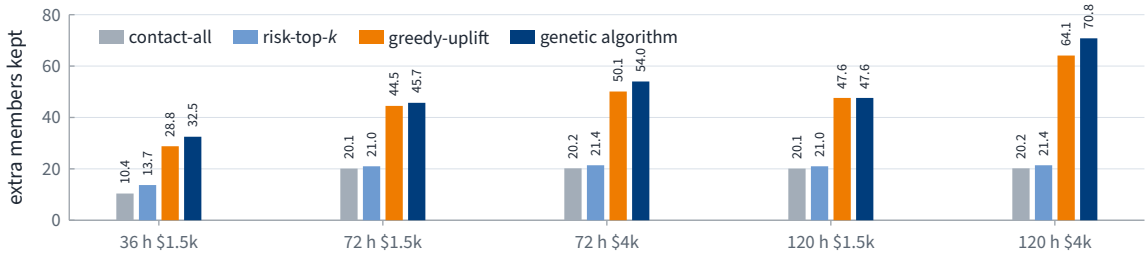


Figure 16 E4 with perfect knowledge: extra members kept by each planner, by staff hours and weekly budget.

Source: `results/e4_optimiser.csv`, mean over three seeds.

routes no better than a keyword router (Figure 20), mostly by answering off-topic questions instead of refusing.

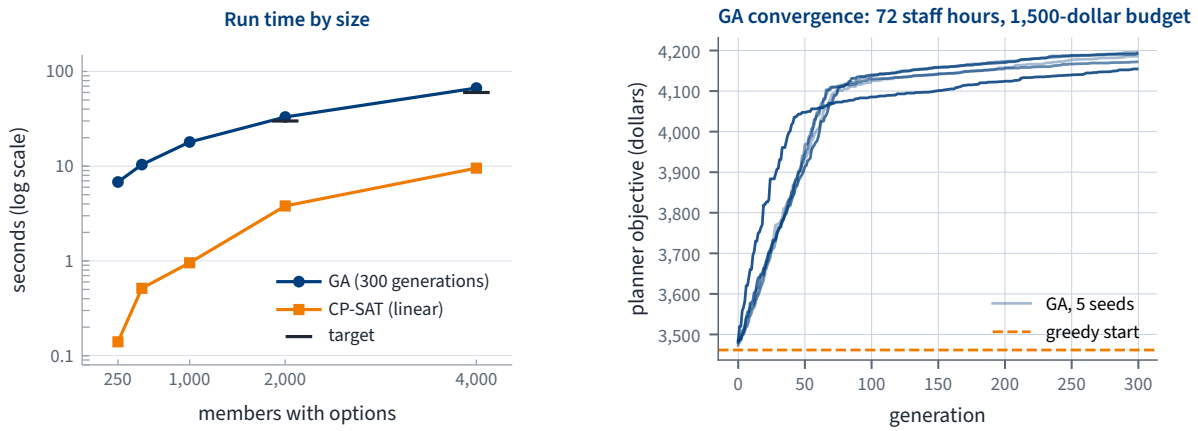


Figure 17 E5. Left: run time grows about linearly; this run missed the 30 s and 60 s targets by about 10%. Right: five seeds climb from the greedy start and plateau about 20% above it (regenerated on the seeded database; the committed run's five seeds end within 0.2% of each other).

Source: `results/e5_speed.csv`; `scripts/figures.py` (ga), as in `b_exact.ipynb`.



Figure 18 E6. Left: of 3,448 proposals over 1,422 members, guardrails remove 1,080 (31.3%), almost all of them contacts inside the 30-day cap. Right: their cost in plan value as the scenario measures it, which has no contact-fatigue effect and so cannot see what the cap protects against.

Source: `results/e6_guardrail_bites.csv`, `results/e6_plan_value.csv`; `c_e6_rules.ipynb`.

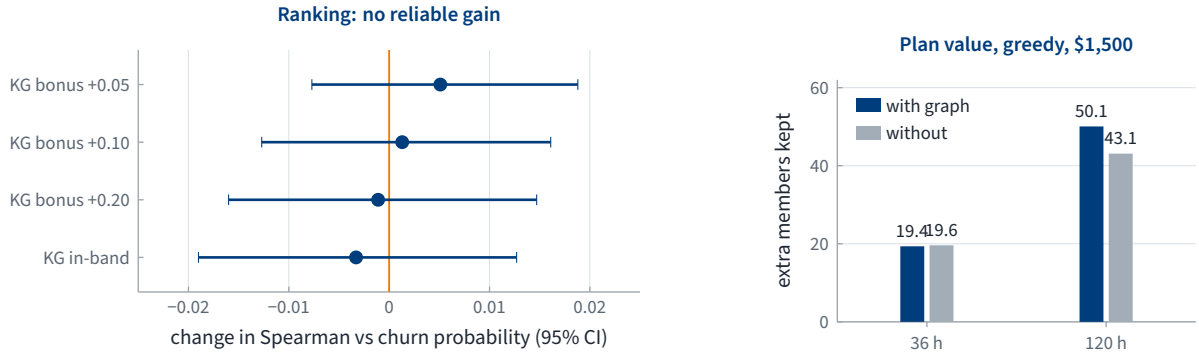
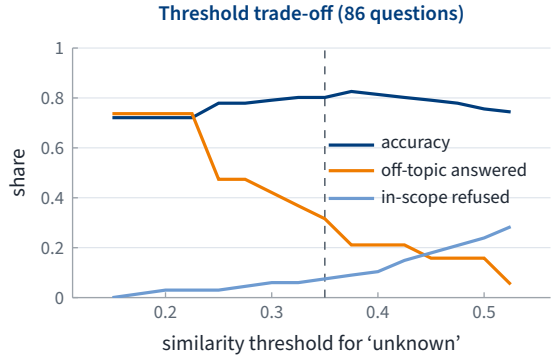


Figure 19 E7. Left: adding the graph's event flags to the churn ranking changes Spearman correlation with the true hazard by less than its uncertainty, whatever the bonus. Right: plan value of the greedy planner, the best in every setting, with and without the graph facts; the gain appears only when staff time is spare.

Source: `results/e7_ranking.csv`, `results/e7_plan_value.csv`; `c_e7_kg.ipynb`.



Questions	n	Classifier	Keywords
AI-written	62	**91.9**	77.4
Held out	24	50.0	**54.2**

Intent accuracy (%).

Figure 20 E8: raising the threshold refuses more off-topic questions but also more in-scope ones; on held-out questions the classifier is no better than keywords.

Source: `results/e8_threshold_sweep.csv`, `results/e8_intent_predictions.csv`.

7 Conclusions

AnchorAI shows that a gym's weekly retention decision can be automated and explained end to end. Real data supports a calibrated churn model and interpretable segments; planning by estimated effect keeps substantially more members than planning by risk; and the rules, graph and assistant add safety, reach and transparency. The limits are as clear: the effects that grade the plans are synthetic, the data is one snapshot, no real gym staff used the system, the assistant's routing fails on unseen questions, and estimate quality, not search, limits the planner. The next steps follow directly: time-stamped data and a small random control group each week to learn real effects; uncertainty-aware planning that stops a strong optimiser amplifying noisy estimates; out-of-scope detection for the assistant; and a field study with gym staff.

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